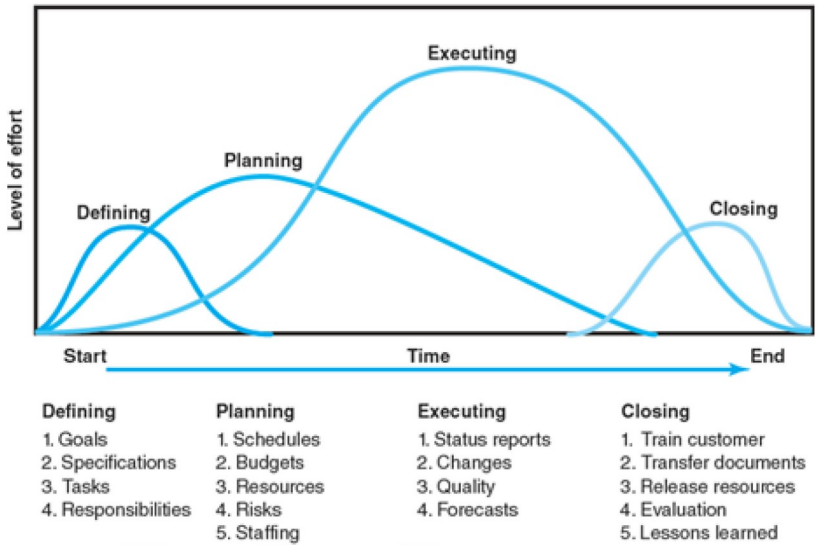
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1. Agile approach and teamwork

In general, the projects are built according to a sequential map as in the following figure, the supervisor of the project is project management whose role is to arrange the planning procedure and take care of daily management and control.



The main principles of conventional project management are roughly as follows:

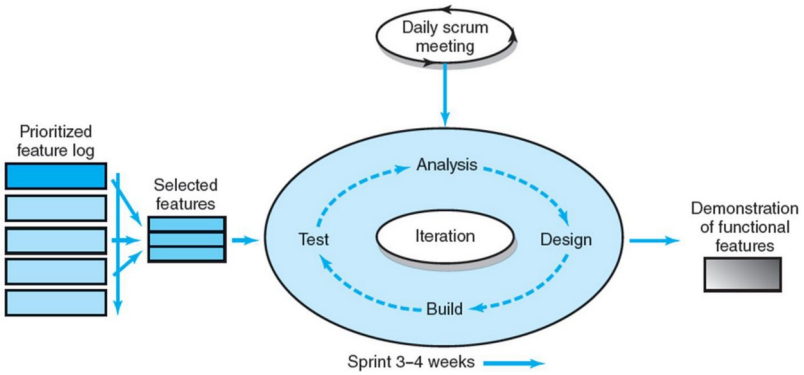
1. Scope the project by defining business case, constraints etc.
2. Define objectives that are quantitative (measurable)
3. Prepare a WBS (Work Breakdown Structure) and align it with the OBS (Organization Structure for the project)
4. Do time based schedule and allocate resources to acquire the Cost Baseline
5. Do a risks assessment
6. Define the monitoring and tracking system e.g. by Earned Value
7. Define Change management system
8. Close down of project

These objectives make up the project cycle and define step by step all the procedures to be implemented for the project to succeed.

In the case we're analysing, we decided to use an agile approach to reduce the deadline of the project and thus reduce the damage caused by mismanagement of the business. In this new scheme, responsibility for scope and schedule trade-off goes to the product owner. Quality management becomes a responsibility

shared among the team, a product owner and Scrum- master. Other traditional tasks are distributed as well among a team's agile project management roles.

The model we propose is based on selecting features that can improve the case by running them according to periodic sprints.
Each sprint is composed of several phases such as testing, analysis, design and construction but behind each phase there are daily meetings between all team members to evaluate, modify or delete any errors.



In this context, the group works together on all aspects of the project with a brainstorm of ideas that are then divided on the board specifying priorities, economic and time forecasts.

In this new approach, project management is replaced by the Scrum-master who is responsible for making sure a Scrum team lives by the values and practices of Scrum. The Scrum-master is often considered a coach for the team, helping the team do the best work it possibly can. The Scrum-master is also often viewed as a protector of the team.

2. Financial and Non-Financial Argumentation

Before highlighting the financial evolution of this case, I want to introduce what is meant by economic and time forecasting and what the most important steps are. The arguments I'm presenting can be taken to compare the design of the case we're looking at and then the design of the business model we want to implement to improve the condition of the company.

Given the urgency to start work on the project, managers sometimes minimize or avoid the effort to follow through on estimating project time and cost. This attitude is a huge mistake and costly. Inaccurate estimates lead to false expectations and consumer dissatisfaction. Ideally, the project manager, and in most cases the customer, would prefer to have a database of detailed schedule and cost estimates for every work package in the project.
Project, people, and external factors all need to be considered to improve quality of estimates for project times and costs, for example, accuracy of estimates depends on the skills of the people making the estimates or external factors such as a pandemic is an external factor that can totally disrupt the plan, an other external factor could be vacations (as in the case we are looking) which can influence project estimates.

There are some general rules to be respect in order to have a good estimation:

1. At the work package level, estimates should be made by the persons most familiar with the task
2. Use several people (or companies) to estimate
3. All task time estimates need consistent time units
4. Estimators should treat each task as independent of other tasks
5. Adding risk assessment to the estimate helps to avoid surprises to stakeholders.

Planning strategies are divided into two blocks: top-down estimates and bottom-up estimates. The first involves planning by the most authoritative people in a company while the second involves those who have direct contact with the work they are doing. The advantages/disadvantages of both procedures are described in the following table.
According to this table, it is evident that the recommended procedure for the case in analysis is that one down-top estimates where there is the necessity to optimize the time and the costs reducing at the same time the uncertainty.

Condition	Top-Down Estimates	Bottom-Up Estimates
Strategic decision making	X	
Cost and time important		X
High uncertainty	X	
Internal, small project	X	
Fixed-price contract		X
Customer wants details		X
Unstable scope	X	

Now let's talk about the case in question, highlighting all the financial steps of the failed project .
The "unsuccessful" goal of the case we are analyzing was to increase by 25% the manufacturing by adding some new machines for the production. This idea was met with mixed opinions, in fact the vice-president and the finance management did not agree for several reasons such as there was already spare land on their property to extend the manufacturing, moving the company means stopping the production and so wasting time and the nearest potential location was 25km further out from the residential area, it disadvantages almost all workers.

Despite this, the idea went ahead, and the first economic and time forecasts began. The internal forecasts estimated (not including office renovation) a cost of €50 million. Ronny (the CEO of the company) agreed with this figure, but the project should not last more than 18 months.

The finance management (without involving the production people into the planning) called an expert Industrial Developers (EID) to quote on the planned expansion which presented a project cost of €80 million that is €30 more than what was supposed to be.

The finance management wanted to reject the EID proposal looking for someone else but in the end, He accepted because EID proposed an hourly rate to pay all the workers involved into the project.

Meanwhile, the controller management developed a monthly cash flow:

a. €5 million = contingencies (before to start)
b. €5 million = the first and the last month
c. €2,3m = each month
d. All the other cost would be recorded as part of the company's normal book-keeping

At the end of the timeframe, the project was only 85% complete and the company applied for an extra loan with a 2% rate trying to complete what they had started.
The results were that the new machines weren’t used, and the production stoppage drove away customers who had previously ordered work. This led to the total failure of the company.

3. Feasibility argumentation

The feasibility of the project we want to present, is based on the analysis of the causes of failure of the project presented in the case.
In fact, it turns out that the main problem of this failure is not the lack of customers or production, but a bad administrative management composed of a familiar and friendly hierarchy that should not exist in a corporate structure.
For example, during project planning, the model down-top estimations were not considered where insiders were not really asked, another mistake was to leave the cost forecasting to a single external company without consulting others.
Yet another, was putting the key to the project such as software management for automated machine work, in the hands of an engineer who has no experience in the field and therefore is not aware of all the bureaucratic and administrative aspects that this type of work requires.
Lastly, there was little transparency on the part of each member underestimating or hiding any work and communication deficiencies, this led to a difficult post-project evaluation and therefore it was difficult to find the causes of failure.
All of the arguments I proposed above will be the pillars that will give life to the project we want to discuss: we strongly believe that by changing administrative roles and planning choices, the failed project can be given a second chance.

3.3 Business case

3.3.1 The purpose of the project, solution and benefits

The purpose of the project is to Increase of 25% the manufacturing adding some new machines for the production of wooden furniture for domestic and industrial use.
These machines need the development and installation of software and hardware which make them work semi-automatically.
Ronny is the CEO of the company and also the one who funds the whole project, Spencer who is already Finance and Administration, becomes project manager and Jon (Ronny's son) runs the management of the company.

3.3.3 Executive project team

The project team consists of 4 people who will actively participate in every project discussion, from brainstorming to planning to fundraising to project post-evaluation. The coordinator is defined scrum master and is one of the members of the team project, his role will be to verify that everyone does their job coordinating the higher authorities of the company during the various meetings. The executive team will base its work on an agile approach in order to reduce delivery times and recover customer confidence as quickly as possible.

3.3.4 Resources and planning

All resources come from the CEO of the company who is willing to pay no more than 50 million, the project has a duration of 18 months from the estimate to the conclusions.
A call for applications will be launched to hire a software engineer, among other requirements there will be two years of experience in the field and ability to work in a team.
Thus, all appropriate agencies will be called in to check for document compatibility regarding all work machines. The project will begin only when each regulation has been verified by the appropriate agency.
Each team member must complete daily questionnaires on the progress of the project and must ensure maximum transparency by highlighting what is going well and what is going wrong.

4. Cost benefits analysis

CBA is a process for evaluating the viability of a particular project in terms of cost and benefit´s where the cost is the value of money used and are no longer available while the benefits is the monetary value of the decisions.
Our cost benefits analysis covers 5 year production and the assumptions we made are:

- Benefit decreasement/year= 4%
- Discount factor=3%
- Cost/incresement/year=5%

First of all, we set the initial investment where almost all the budget is divided by the building of the train production and the space (house) for it.

New equipment			
Initial investment	Quantity	Cost/item	Total
new train production mach	1	20.000.000	20.000.000
house for the new product	1	20.000.000	20.000.000
machine licenses	1	1.000.000	1.000.000
audit (machines/house)	2	1.000.000	2.000.000
uncertainty	1	5.000.000	5.000.000
	0	-	-
	0	-	-
			48.000.000

Then, we set the cost for year that this project involves, the most important figure is the software engineer who will care of the core of the train production also after the project is over

Cost per year	Quantity	Cost/item	
machines manatemainten	2	70.000	140.000
Software engineer	1	170.000	170.000
Facility cost	1	3.000	3.000
Consultancy	1	700	700
			313.700

We calculated an additional benefit of 25% due to the expansion of the production train, this brings a benefit of about 20 million

Benefits per first year	
Initial investment	Cost/item
outcomes from the production	20.000.000

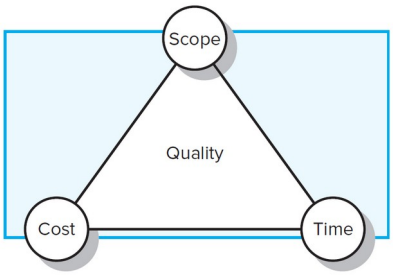
Finally, we come out with the NPV value which is a positive number of about 13, we have the year 0 which represents the initial investment and then 4 years of production.

Year		0		1		2		3		4		5
Initial investment	-	48.000.000										
Cost per year			-	313.700	-	329.385	-	345.854	-	363.147	-	381.304
Cost per year	-	48.000.000	-	313.700	-	329.385	-	345.854	-	363.147	-	381.304
Benefits per year		0		20.084.000		19.280.640		18.509.414		17.769.038		17.058.276
Scrap value												12.000
Net benefits:	-	48.000.000		19.770.300		18.951.255		18.163.560		17.405.891		16.688.972
	-	48.000.000		17.191.565		14.329.871		11.942.836		9.951.875		8.297.369
NPV=		13.713.516										

5. Scope the project

Our scope is to increase the production of wood furniture to 25 % according to the budget of the CEO Mr Rooney, who can deliver it.
The project is 18 months long and it needs 50 million as investment.
We will deliver a good software for the semi-automatically train production of wood furniture, this new

system will increase the number of customers in time.
The new train production machine requires more space, that is why we have to take into account to enlarge the production space by constructing a new building.
Technical requirements will have to evaluate the compatibility of new machines with current regulations so that production is not blocked.
The team project has the duty to check day by day all the progress and the outcomes according to the following scheme:



6. SMART

In order to reach the main concerns a project may have, our project must be SMART.
SMART is an acronym whose letters represent several features; so let's examine them one by one showing why out project is smart:

1. Specific: increase the production of wood furniture to 25 %
2. Measurable: we can count how many furniture the company products and its benefits
3. Achievable: According to the budget of 50m, we calculated we have all the resources to make the project successful
4. Realistic: the budget comes from the CEO of the company who has an high credibility to deliver this amount of money and we gave a realist time-line for all the step.
5. Time-bound: the project is 18 months long and the finance planning is 5 years long.

7. WBS & OBS

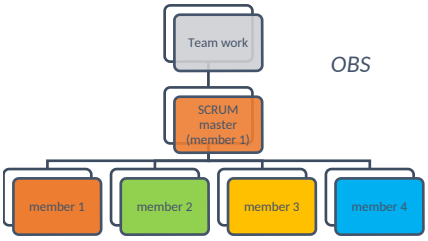
The WBS is a view into the project which shows what work the project encompasses. It is a tool which helps to easily communicate the work and processes involved to execute the project. The Project Manager and project team use the WBS to develop the project schedule, resource requirements and costs.

The Organization Breakdown Structure groups together similar project activities or “work packages” and relates them to the organization’s structure. It is used to define the responsibilities for project management, cost reporting, billing, budgeting and project control. The OBS provides an organizational rather than a task-based perspective of the project. The hierarchical structure of the OBS allows the aggregation (rollup) of project information to higher levels. When project responsibilities are defined and work is assigned, the OBS and WBS are connected providing the possibility for powerful analytics to measure project and workforce performance at a very high level (example business unit performance) or down to the details (example user work on a task).
To develop an Organization Breakdown Structure, we need to:

1. Draw of the entire organization as a hierarchy
2. Define all departments and project teams.
    1. Specify functional (where cost for the work the user does is allocated to) and approval (who approves the work the user performs and any leave time approvals) groups for every user.

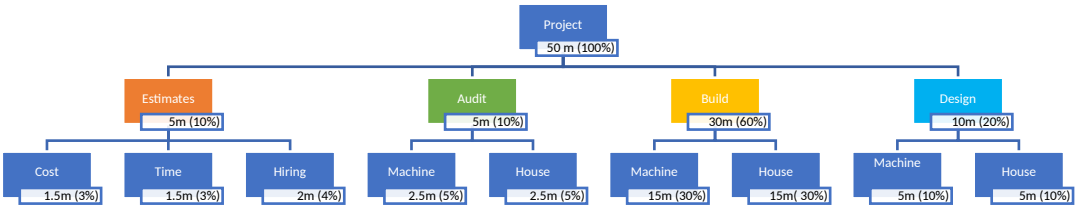
The WBS and OBS we have built are linked in the following structure where each colour of the member represents their area of interest. For simplicity we have not introduced in the scheme all the subfigures

related to houses and machines, but we list them now: construction workers, electronic, civil and automation engineers.



OBS

WBS



9. Business case Alternative

We also thought an alternative project idea in case the project still doesn't work, It might be a good start-point to make the company great again.

9.1 The purpose of the project, solution and benefits

The purpose of the business case project is to recover the company from bankruptcy by investing an additional sum of money to complete the work already begun but at the same time change the managerial roles and planning model used. The main problem we want to solve is to recover the trust of customers who have seen delays in production.
To do this we thought of an advertising campaign that highlights the automated software (introduced in the new project) with the possible possibility of selling this system to other companies that need it. In this way we add an additional economic income that can make us recover in time all the money invested in the project in a faster way. The software will have the same name as the company and will become a mark of quality, speed and optimization of industrial work.

9.2 Executive project team

The project team consists of 4 people who will actively participate in every project discussion, from brainstorming to planning to fundraising to project post-evaluation. The coordinator is defined scrum master and is one of the members of the team project, his role will be to verify that everyone does their job coordinating the higher authorities of the company during the various meetings. The executive team will base its work on an agile approach in order to reduce delivery times and recover customer confidence as quickly as possible.

9.3 Resources and planning

The resources will be reached by publicizing the business idea and looking for possible shareholders willing to believe in the project, the company's assets will be divided into stokes. We will propose another corporate structure proposing to dismiss all the figures that led to the previous failure.
Project planning depends mainly on the time it takes to find the resources, we think that the project can start only if all the planned resources are reached. In that case we think that a first sprint can take €50m

which will be used to complete 15% of the work left unfinished, optimize even more the software and launch an advertising campaign.